

MQTT – Modular Controller Board

Summary:

ESP32 S3 based controller for attached modules. Designed to receive commands via MQTT and send required information. This version supports four individual modules being attached. USB C connection for programming, and only supports the controller itself.

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Non BOM Items

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H101
MountingHole 3.5mm
- 

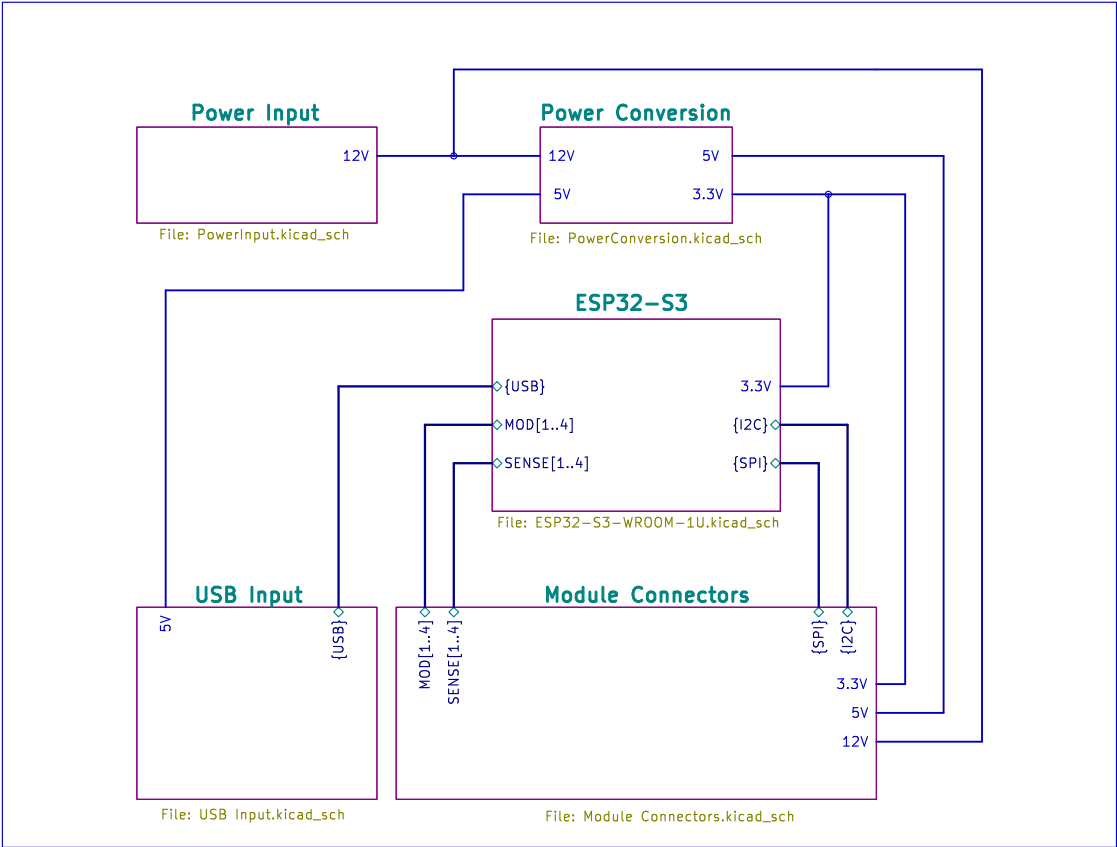
H103
MountingHole 3.5mm
- 

H102
MountingHole 3.5mm
- 

H104
MountingHole 3.5mm

Additional BOM Items

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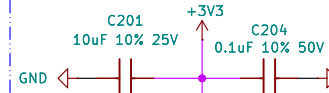


Sheet: /		
File: MQTT-ModularControllerBoard.kicad_sch		
Title: MQTT – Modular Controller Board		
Size: A4	Date: 2026-03-14	Rev: 1.0
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SENSE1
SENSE2
SENSE3
SENSE4

SENSE[1..4]

Decoupling capacitors per datasheet.



MOD1
MOD2
MOD3
MOD4

MOD[1..4]



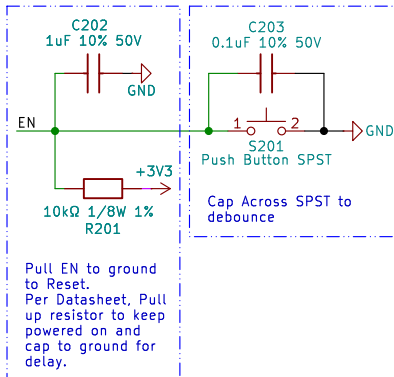
N16R8 Variant uses these pins to communicate with internal PSRAM

GND

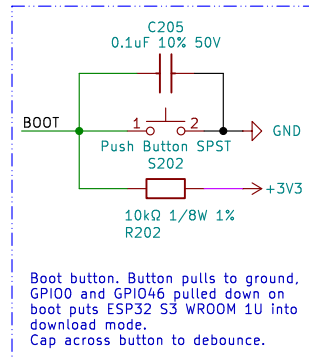
GND

Strapping pins pulled low to enforce default behaviour.

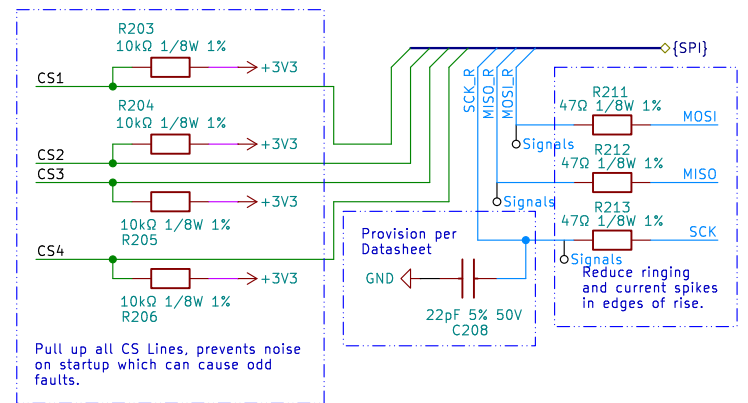
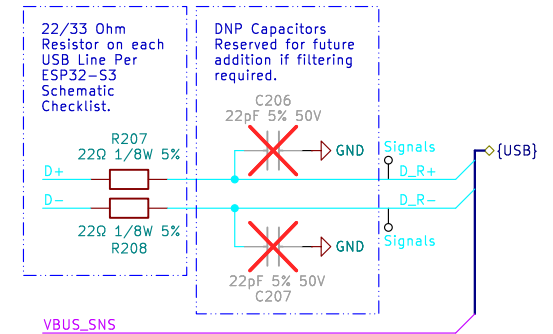
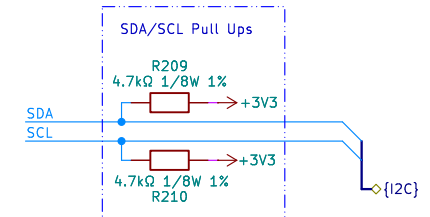
- GPIO3 Sets JTAG Mode, Enforce Default, not being used.
- GPIO45 Controls Voltage to Package RAM, enforce 3.3V
- GPIO46 Low Enters Download Mode if GPIO0 (Boot) is High, Enforce low, controlled by GPIO0.



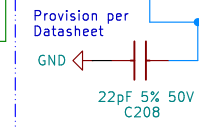
Pull EN to ground to Reset. Per Datasheet, Pull up resistor to keep powered on and cap to ground for delay.



Boot button, Button pulls to ground, GPIO0 and GPIO46 pulled down on boot puts ESP32 S3 WROOM 1U into download mode. Cap across button to debounce.



Pull up all CS Lines, prevents noise on startup which can cause odd faults.



Reduce ringing and current spikes in edges of rise.

Summary:

- USB connection for programming, USB VBUS SENSE on GPIO08
- Manual backup writing, hold boot and press reset, then release boot to put in download mode.
- SDA/SCL & SPI for communication with Modules, Module CS Pins, GPIO06, GPIO07, GPIO15, GPIO16
- One General use pin per module, GPIO40, GPIO42, GPIO43, GPIO01
- One Detection Pin, GPIO39, GPIO41, GPIO44, GPIO02

Sheet: /ESP32-S3/
File: ESP32-S3-WROOM-1U.kicad_sch

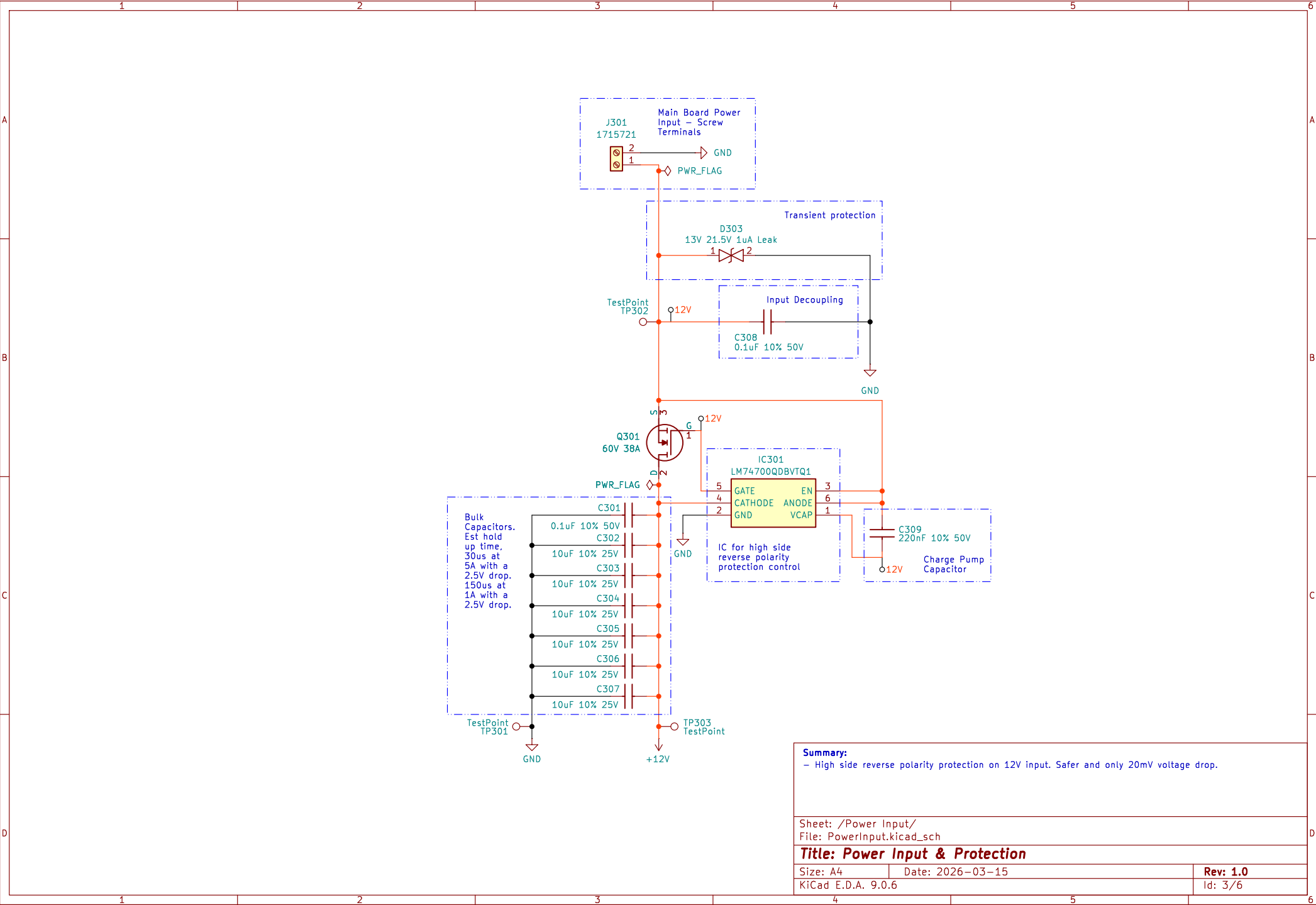
Title: ESP32 S3 WROOM 1U Controller

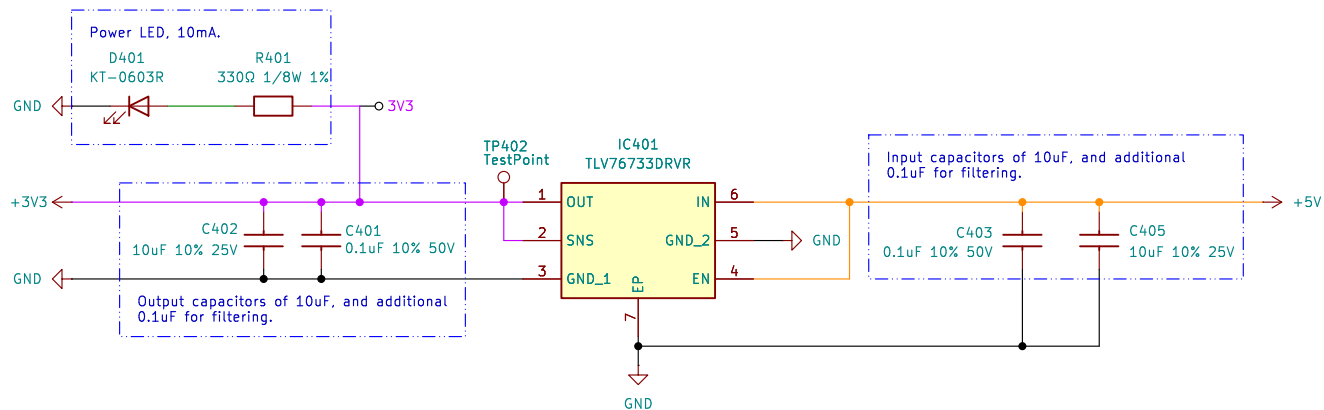
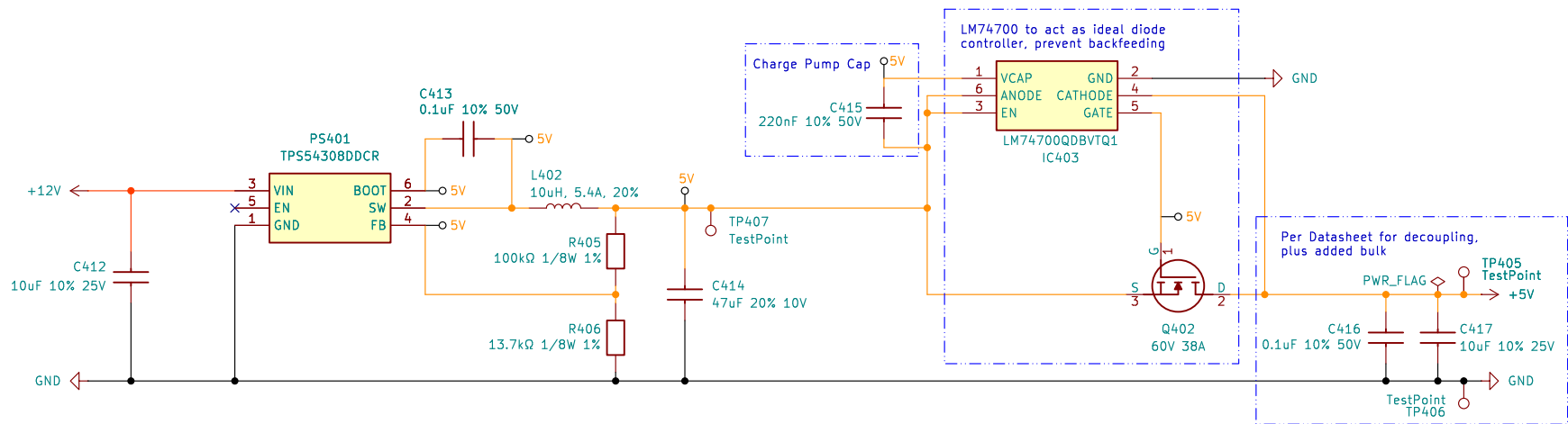
Size: A4 Date: 2026-01-18

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Summary:

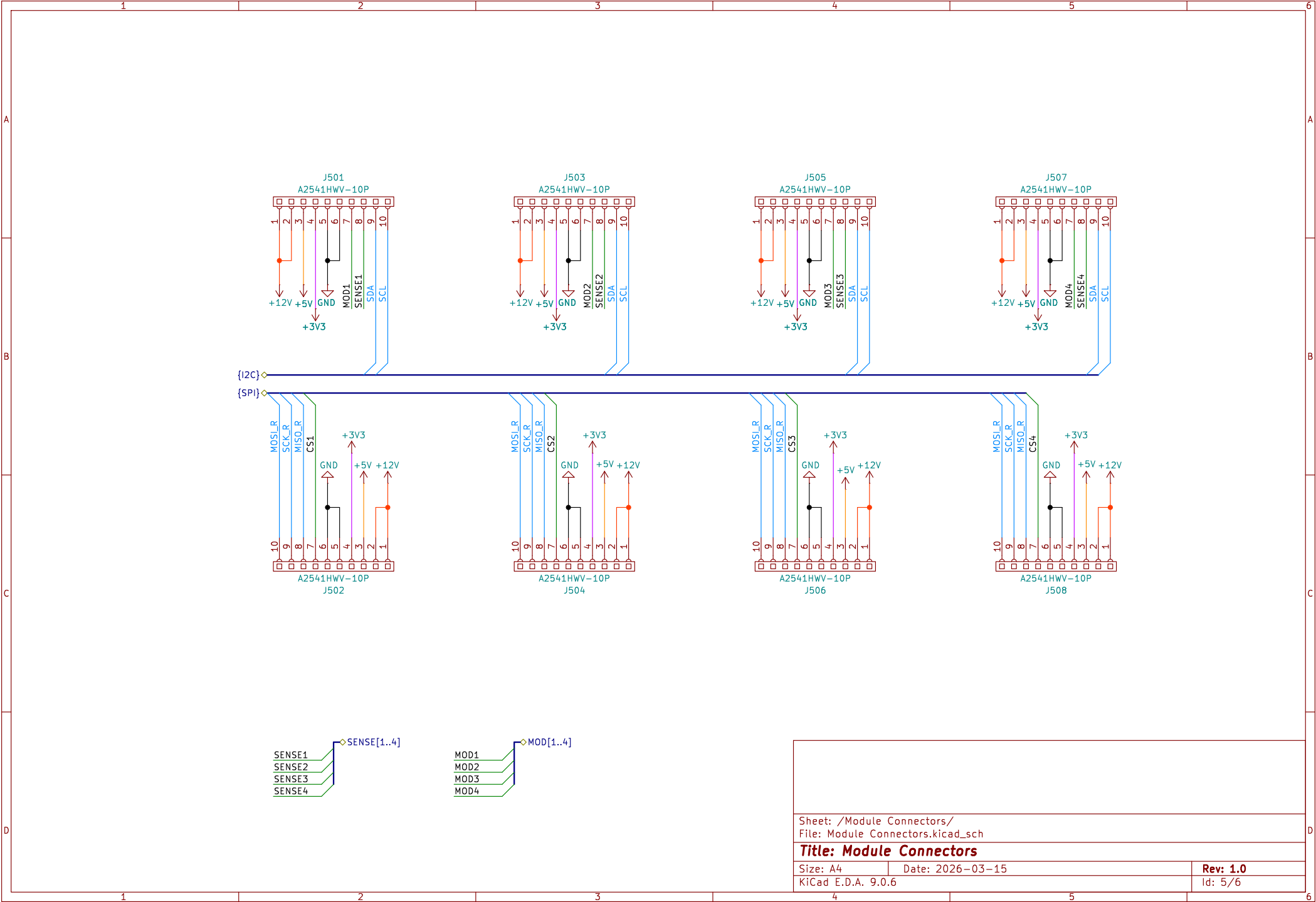
- Buck converter generated with TI WEBENCH POWER DESIGNER for 5V output and max 3A output.
- LDO Conversion of 5V Power to 3.3V with a max Peak of 1A. Ideally < 800mA.

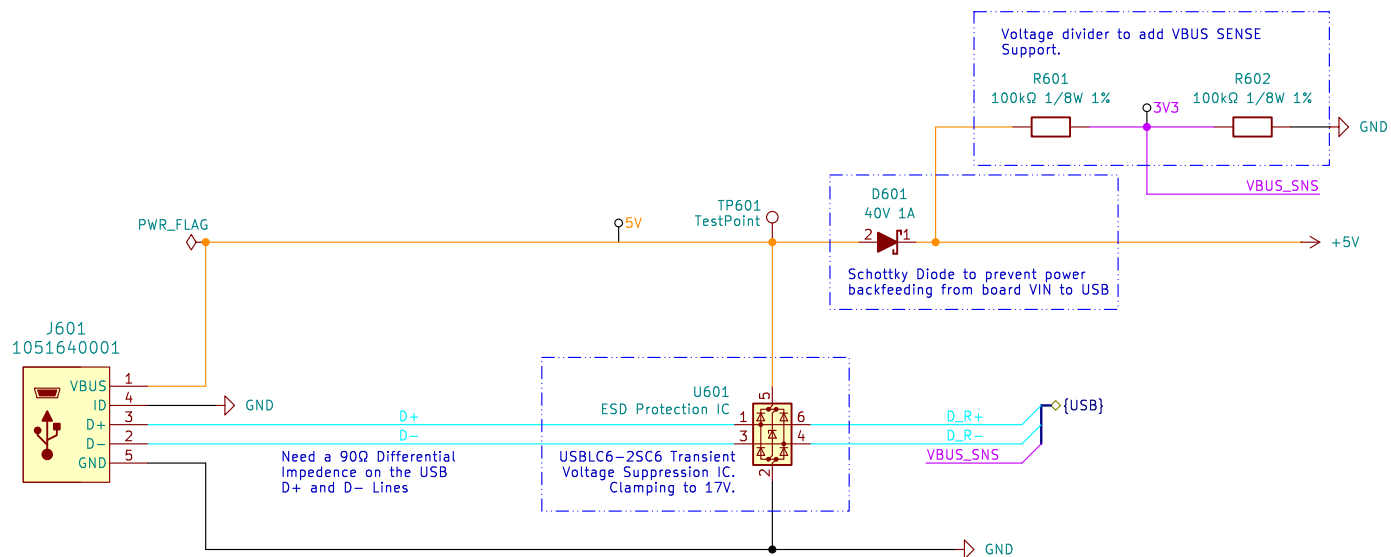
Sheet: /Power Conversion/
File: PowerConversion.kicad_sch

Title: Power Conversion

Size: A4 Date: 2026-07-24
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Id: 4/6





Summary:
 - VBUS_SNS in place for USB connection while connected to 12V. Requires detection of being self powered on startup in software.

Sheet: /USB Input/
 File: USB Input.kicad_sch

Title: USB C Connection

Size: A4 Date: 2026-03-18

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